NLP LAB PROGRAMS

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Program 1

Programs on text tokenization, stop word removal, stemming/lemmatization.

THEORY:

Tokenization: Process of splitting text into smaller units like words or sentences.

Stopword Removal: Eliminating common words (e.g., “is”, “the”) that do not add meaningful information.

Stemming: Reducing words to their root form by removing suffixes (may not be linguistically correct).

Lemmatization: Converting words to their base/dictionary form using vocabulary and grammar rules.

CODE:

```
import nltk

nltk.download('punkt_tab')

from nltk.tokenize import word_tokenize

from nltk.corpus import stopwords

from nltk.stem import PorterStemmer, WordNetLemmatizer


# Sample text

text = "Natural language processing includes tokenization, removing stopwords, stemming, and lemmatization."


# 1. Tokenization

tokens = word_tokenize(text)

print("Tokens:", tokens)


# 2. Stop word removal

stop_words = set(stopwords.words('english'))
```

```
filtered_tokens = [word for word in tokens if word.lower() not in stop_words]
print("After Stop Word Removal:", filtered_tokens)

# 3. Stemming
ps = PorterStemmer()
stemmed = [ps.stem(word) for word in filtered_tokens]
print("Stemmed:", stemmed)


# 4. Lemmatization
lemmatizer = WordNetLemmatizer()
lemmatized = [lemmatizer.lemmatize(word) for word in filtered_tokens]
print("Lemmatized:", lemmatized)
```

OUTPUT:

```
Tokens: ['Natural', 'language', 'processing', 'includes', 'tokenization', ',', 'removing', 'stopwords', ',', 'stemming', ',', 'and', 'lemmatization',
'.']
After Stop Word Removal: ['Natural', 'language', 'processing', 'includes', 'tokenization', ',', 'removing', 'stopwords', ',', 'stemming', ',', 'lemmatiza
tion', '.']
Stemmed: ['natur', 'languag', 'process', 'includ', 'token', ',', 'remov', 'stopword', ',', 'stem', ',', 'lemmat', '.']
Lemmatized: ['Natural', 'language', 'processing', 'includes', 'tokenization', ',', 'removing', 'stopwords', ',', 'stemming', ',', 'lemmatization', '.']
```

Program 2

Analyze the pre-processed data and visualize the effects of cleaning.

THEORY:

Text Cleaning: Removing noise such as punctuation, stopwords, and non-alphabetic tokens to improve data quality.

Normalization: Standardizing text (e.g., lowercase conversion) for consistent processing.

CODE:

```
import nltk

from nltk.tokenize import word_tokenize

from nltk.corpus import stopwords

from nltk.stem import PorterStemmer, WordNetLemmatizer


# Download required NLTK resources

nltk.download('punkt')

nltk.download('stopwords')

nltk.download('wordnet')

nltk.download('omw-1.4')


text="""Natural language processing includes tokenization, removing stopwords, stemming, and lemmatization.

It helps machines understand human language effectively."""


# Tokenization

tokens=word_tokenize(text)


# Remove stopwords and punctuation
```

```
stop_words=set(stopwords.words('english'))
clean_tokens=[w for w in tokens if w.lower() not in stop_words and w.isalpha()]

# Stemming and Lemmatization
ps=PorterStemmer()
lemmatizer=WordNetLemmatizer()

print("\nWord Transformations:\n")
print(f"{'Original':<20} {'Stemmed':<20} {'Lemmatized'}")
print("-"*55)

for word in clean_tokens:
    print(f"{word:<20} {ps.stem(word):<20} {lemmatizer.lemmatize(word)}")
```

OUTPUT:

```
Word Transformations:

Original             Stemmed              Lemmatized
-------------------------------------------------------
Natural              natur                Natural
language             languag              language
processing           process              processing
includes             includ               includes
tokenization         token                tokenization
removing             remov                removing
stopwords            stopword             stopwords
stemming             stem                 stemming
lemmatization        lemmat               lemmatization
helps                help                 help
machines             machin               machine
understand           understand           understand
human                human                human
language             languag              language
effectively          effect               effectively
```

Program 3

Tokenize It Right: Implement different tokenization techniques on various text samples. Evaluate their effectiveness and choose the best option for a specific task, like sentiment analysis.

THEORY:

Whitespace Tokenization: Splits text based on spaces without handling punctuation.

Rule-based Tokenization (Punkt): Uses linguistic rules to split words and punctuation accurately.

Tweet Tokenization: Specialized tokenizer handling emojis, hashtags, and informal text.

CODE:

```
import nltk

from nltk.tokenize import word_tokenize, WhitespaceTokenizer, TweetTokenizer


# Download NLTK resources (run once)

nltk.download('punkt')

nltk.download('stopwords')


texts=[

    "I love this product! It's awesome :) #happy",

    "Well... I'm not sure if this is good or bad. :/",

    "Ugh, the movie was terrible!!! Worst ever :(",

    "Can't wait for the weekend!!! #excited",

]


ws_tokenizer=WhitespaceTokenizer()

tweet_tokenizer=TweetTokenizer()
```

```
for i,text in enumerate(texts,1):
    print(f"\n--- Text {i} ---\nOriginal: {text}\n")
    print("Whitespace Tokenizer:",ws_tokenizer.tokenize(text))
    print("NLTK word_tokenize:",word_tokenize(text))
    print("NLTK TweetTokenizer:",tweet_tokenizer.tokenize(text))
```

OUTPUT:

```
--- Text 1 ---
Original: I love this product! It's awesome :) #happy

Whitespace Tokenizer: ['I', 'love', 'this', 'product!', "It's", 'awesome', ':)', '#happy']
NLTK word_tokenize: ['I', 'love', 'this', 'product', '!', 'It', "'s", 'awesome', ':', ')', '#', 'happy']
NLTK TweetTokenizer: ['I', 'love', 'this', 'product', '!', "It's", 'awesome', ':)', '#happy']

--- Text 2 ---
Original: Well... I'm not sure if this is good or bad. :/

Whitespace Tokenizer: ['Well...', "I'm", 'not', 'sure', 'if', 'this', 'is', 'good', 'or', 'bad.', ':/']
NLTK word_tokenize: ['Well', '...', 'I', "'m", 'not', 'sure', 'if', 'this', 'is', 'good', 'or', 'bad', '.', ':', '/']
NLTK TweetTokenizer: ['Well', '...', "I'm", 'not', 'sure', 'if', 'this', 'is', 'good', 'or', 'bad', '.', ':/']

--- Text 3 ---
Original: Ugh, the movie was terrible!!! Worst ever :(

Whitespace Tokenizer: ['Ugh,', 'the', 'movie', 'was', 'terrible!!!', 'Worst', 'ever', ':(']
NLTK word_tokenize: ['Ugh', ',', 'the', 'movie', 'was', 'terrible', '!', '!', '!', 'Worst', 'ever', ':', '(']
NLTK TweetTokenizer: ['Ugh', ',', 'the', 'movie', 'was', 'terrible', '!', '!', '!', 'Worst', 'ever', ':(']

--- Text 4 ---
Original: Can't wait for the weekend!!! #excited

Whitespace Tokenizer: ["Can't", 'wait', 'for', 'the', 'weekend!!!', '#excited']
NLTK word_tokenize: ['Ca', "n't", 'wait', 'for', 'the', 'weekend', '!', '!', '!', '#', 'excited']
NLTK TweetTokenizer: ["Can't", 'wait', 'for', 'the', 'weekend', '!', '!', '!', '#excited']
```

Program 4

Stemming Showdown: Apply stemming and lemmatization to a movie review corpus and compare their outputs. Analyze their impact on tasks like information retrieval or stemming accuracy.

THEORY:

Stemming vs Lemmatization Comparison: Evaluates trade-off between speed (stemming) and accuracy (lemmatization).

Information Retrieval Impact: Text normalization affects search accuracy and matching efficiency

CODE:

```
import nltk

from nltk.tokenize import word_tokenize

from nltk.stem import PorterStemmer, WordNetLemmatizer


# Download required resources once

nltk.download('punkt')

nltk.download('wordnet')

nltk.download('omw-1.4')


# Sample movie reviews (small corpus)

reviews=[

    "The movie was fantastic! I loved the thrilling scenes and amazing acting.",

    "It was boring and too long. I didn't like the predictable plot.",

    "Absolutely wonderful cinematography and great story.",

]
```

```
ps=PorterStemmer()
lemmatizer=WordNetLemmatizer()

for i,review in enumerate(reviews,1):
    print(f"\n--- Review {i} ---\nOriginal:",review)
    tokens=word_tokenize(review)
    print("Stemmed:   ",' '.join(ps.stem(word) for word in tokens))
    print("Lemmatized:",' '.join(lemmatizer.lemmatize(word) for word in tokens))
```

OUTPUT:

```
--- Review 1 ---
Original: The movie was fantastic! I loved the thrilling scenes and amazing acting.
Stemmed:    the movi wa fantast ! i love the thrill scene and amaz act .
Lemmatized: The movie wa fantastic ! I loved the thrilling scene and amazing acting .

--- Review 2 ---
Original: It was boring and too long. I didn't like the predictable plot.
Stemmed:    it wa bore and too long . i did n't like the predict plot .
Lemmatized: It wa boring and too long . I did n't like the predictable plot .

--- Review 3 ---
Original: Absolutely wonderful cinematography and great story.
Stemmed:    absolut wonder cinematographi and great stori .
Lemmatized: Absolutely wonderful cinematography and great story .
```

Program 5

NER in Action: Develop a Named Entity Recognition system to identify entities in news articles for one category (people, locations, or organizations). Evaluate its performance using standard metrics.

THEORY:

NER: Identifies and classifies entities like persons, locations, or organizations in text.

Evaluation Metrics: Precision, recall, and F1-score measure model performance.

CODE:

```
import nltk
from nltk import word_tokenize, pos_tag, ne_chunk
from nltk.tree import Tree
from sklearn.metrics import precision_recall_fscore_support
nltk.download('maxent_ne_chunker')
nltk.download('words')

# Sample data: list of (sentence, true_entities)
# true_entities are PERSON names manually labeled as lists of strings
data=[
    ("Barack Obama was born in Hawaii.", ["Barack Obama"]),
    ("Apple is looking at buying U.K. startup for $1 billion.", []),
    ("Elon Musk is the CEO of SpaceX.", ["Elon Musk"]),
    ("Angela Merkel met with Vladimir Putin in Moscow.", ["Angela Merkel", "Vladimir
Putin"]),
]
def extract_entities(text, label="PERSON"):
    tokens=word_tokenize(text)
```

```
    chunks=ne_chunk(pos_tag(tokens))
    return [" ".join(c[0] for c in chunk.leaves()) for chunk in chunks if type(chunk)==Tree and chunk.label()==label]

# Extract predicted PERSON entities
y_true=[]
y_pred=[]

for sent,true_entities in data:
    pred_entities=extract_entities(sent,"PERSON")
    y_true.append(set(true_entities))
    y_pred.append(set(pred_entities))

# Flatten sets into lists of all entities (for simple evaluation)
true_all=[ent for ents in y_true for ent in ents]
pred_all=[ent for ents in y_pred for ent in ents]
all_entities=list(set(true_all+pred_all))

# Create binary lists for precision/recall calculation (entity-level)
def binary_labels(entity_list):
    return [1 if ent in entity_list else 0 for ent in all_entities]

y_true_bin=[binary_labels(ents) for ents in y_true]
y_pred_bin=[binary_labels(ents) for ents in y_pred]

# Average across sentences
import numpy as np
y_true_bin_avg=np.mean(y_true_bin,axis=0)
```

```
y_pred_bin_avg=np.mean(y_pred_bin,axis=0)
# For simplicity, calculate precision/recall/F1 across all predicted entities
precision,recall,f1,_=precision_recall_fscore_support(
    [1]*len(all_entities),
    [1 if ent in pred_all else 0 for ent in all_entities],
    average='micro'
)

print("Extracted PERSON entities per sentence:")
for i,(sent,true_e) in enumerate(data):
    print(f"Sentence: {sent}")
    print(f"True Entities: {true_e}")
    print(f"Predicted Entities: {y_pred[i]}\n")

print(f"Precision: {precision:.2f}")
print(f"Recall:    {recall:.2f}")
print(f"F1-score:  {f1:.2f}")
```

OUTPUT:

```
Extracted PERSON entities per sentence:
Sentence: Barack Obama was born in Hawaii.
True Entities: ['Barack Obama']
Predicted Entities: {'Barack', 'Obama'}

Sentence: Apple is looking at buying U.K. startup for $1 billion.
True Entities: []
Predicted Entities: set()

Sentence: Elon Musk is the CEO of SpaceX.
True Entities: ['Elon Musk']
Predicted Entities: {'Elon'}

Sentence: Angela Merkel met with Vladimir Putin in Moscow.
True Entities: ['Angela Merkel', 'Vladimir Putin']
Predicted Entities: {'Vladimir Putin', 'Angela', 'Merkel'}

Precision: 0.67
Recall:    0.67
F1-score:  0.67
```

Program 6

N-gram Language Prediction: Build and compare unigram, bigram, and trigram language models for a short story corpus. Evaluate their next word prediction accuracy and discuss limitations

THEORY:

N-gram Model: Predicts next word based on previous 'n' words using probability.

Language Modeling: Estimates likelihood of word sequences in text.

CODE:

```
import re

from collections import defaultdict


# Sample short story

text="""

Once upon a time, there was a clever fox who lived in a forest.

The fox was hungry and looking for food.

He saw a crow sitting on a tree with a piece of cheese in her beak.

The fox thought of a clever plan to get the cheese.

He praised the crow's voice and asked her to sing.

As she opened her beak, the cheese fell down, and the fox quickly ate it.

"""


# Step 1: Preprocess text

def preprocess(text):

    return re.sub(r'[^\w\s]','',text.lower()).split()
```

```
# Step 2: Build N-gram model
def build_model(words,n):
    model=defaultdict(lambda:defaultdict(int))
    for i in range(len(words)-n):
        model[tuple(words[i:i+n])][words[i+n]]+=1
    return model

# Step 3: Predict next word
def predict(model,context):
    context=tuple(context)
    return max(model[context],key=model[context].get) if context in model else None

# Step 4: Evaluate accuracy
def evaluate(model,words,n):
    total=len(words)-n
    correct=sum(predict(model,words[i:i+n])==words[i+n] for i in range(total))
    return correct/total if total else 0

# Prepare the models
words=preprocess(text)

# Interactive Menu
def interactive():
    print("Welcome to the N-gram Predictor!")
    print("Sample Text:\n",text.strip())
    while True:
        n_input=input("\nChoose N-gram size (1 = Unigram, 2 = Bigram, 3 = Trigram, or type
'exit' to quit): ")
```

```
    if n_input.lower()=="exit":
        print("Goodbye!")
        break
    if not n_input.isdigit() or int(n_input)<1:
        print("Please enter a valid number greater than 0.")
        continue
    n=int(n_input)
    model=build_model(words,n)
    print(f"\nModel built using {n}-grams. Accuracy: {evaluate(model,words,n):.2f}")

    while True:
        user_input=input(f"\nEnter {n} word(s) as context (or 'back' to change N-gram size): ").lower()
        if user_input=="back":
            break
        if user_input=="exit":
            print("Goodbye!")
            return
        context_words=user_input.strip().split()
        if len(context_words)!=n:
            print(f"Please enter exactly {n} word(s).")
            continue
        next_word=predict(model,context_words)
        print(f"Predicted next word: {next_word}" if next_word else "No prediction found. Try a different context.")

# Run the interactive session
interactive()
```

OUTPUT:

```
Welcome to the N-gram Predictor!
Sample Text:
 Once upon a time, there was a clever fox who lived in a forest.
The fox was hungry and looking for food.
He saw a crow sitting on a tree with a piece of cheese in her beak.
The fox thought of a clever plan to get the cheese.
He praised the crow's voice and asked her to sing.
As she opened her beak, the cheese fell down, and the fox quickly ate it.

Choose N-gram size (1 = Unigram, 2 = Bigram, 3 = Trigram, or type 'exit' to quit):  1

Model built using 1-grams. Accuracy: 0.70

Enter 1 word(s) as context (or 'back' to change N-gram size):  once
Predicted next word: upon

Enter 1 word(s) as context (or 'back' to change N-gram size):  back

Choose N-gram size (1 = Unigram, 2 = Bigram, 3 = Trigram, or type 'exit' to quit):  2

Model built using 2-grams. Accuracy: 0.93

Enter 2 word(s) as context (or 'back' to change N-gram size):  once upon
Predicted next word: a

Enter 2 word(s) as context (or 'back' to change N-gram size):  back

Choose N-gram size (1 = Unigram, 2 = Bigram, 3 = Trigram, or type 'exit' to quit):  3

Model built using 3-grams. Accuracy: 0.99

Enter 3 word(s) as context (or 'back' to change N-gram size):  once upon a
Predicted next word: time

Enter 3 word(s) as context (or 'back' to change N-gram size):  back

Choose N-gram size (1 = Unigram, 2 = Bigram, 3 = Trigram, or type 'exit' to quit):  exit
Goodbye!
```

Program 7

Markov Chain Text Generation: Implement a Markov chain to generate 10 short poems based on a specific poet's work. Analyze the coherence and resemblance to the original style.

THEORY:

Markov Chain: Probabilistic model where next state depends only on current state.

Text Generation: Produces new text by learning transition probabilities between words.

CODE:

```
import random
from collections import defaultdict

# Emily Dickinson text (short sample; replace with full corpus for better results)
text="""
Because I could not stop for Death –
He kindly stopped for me –
The Carriage held but just Ourselves –
And Immortality.
We slowly drove – He knew no haste
And I had put away
My labor and my leisure too,
For His Civility –
We passed the School, where Children strove
At Recess – in the Ring –
We passed the Fields of Gazing Grain –
We passed the Setting Sun –
"""
```

```
def build_markov_chain(text,order=2):
    words=text.replace("–","").replace("\n"," ").split()
    markov_chain=defaultdict(list)
    for i in range(len(words)-order):
        markov_chain[tuple(words[i:i+order])].append(words[i+order])
    return markov_chain,set(words)

def generate_poem(chain,order=2,max_words=20,line_len=4):
    poem=list(random.choice(list(chain.keys())))
    for _ in range(max_words-order):
        prefix=tuple(poem[-order:])
        if prefix not in chain: break
        poem.append(random.choice(chain[prefix]))
    return "\n".join(" ".join(poem[i:i+line_len]) for i in range(0,len(poem),line_len)),poem

def calculate_resemblance(poem_words,corpus_words):
    poem_set=set(poem_words)
    lexical_score=len(poem_set.intersection(corpus_words))/len(poem_set) if poem_set else 0

poetic_vocab={"Death","Immortality","Soul","Heaven","Carriage","Civility","Sun","Grain",
"Ring"}
    poetic_score=len(poem_set.intersection(poetic_vocab))/len(poem_set)
    return round((0.7*lexical_score+0.3*poetic_score)*100,2)

# Build the chain and get corpus words
order=2
chain,corpus_words=build_markov_chain(text,order=order)
```

```
# Generate poems with resemblance
print("Generated Emily Dickinson-Inspired Poems with Resemblance Score:\n")
for i in range(10):
    poem_text,poem_words=generate_poem(chain,order=order,max_words=16,line_len=4)
    print(f"Poem {i+1} (Resemblance: {calculate_resemblance(poem_words,corpus_words)}%):\n{poem_text}\n")
```

OUTPUT:

```
Generated Emily Dickinson-Inspired Poems with Resemblance Score:

Poem 1 (Resemblance: 76.67%):
His Civility We passed
the Fields of Gazing
Grain We passed the
Fields of Gazing Grain

Poem 2 (Resemblance: 76.67%):
the Fields of Gazing
Grain We passed the
Setting Sun

Poem 3 (Resemblance: 70.0%):
He knew no haste
And I had put
away My labor and
my leisure too, For

Poem 4 (Resemblance: 70.0%):
And Immortality. We slowly
drove He knew no
haste And I had
put away My labor

Poem 5 (Resemblance: 71.88%):
And I had put
away My labor and
my leisure too, For
His Civility We passed
```

```
Poem 6 (Resemblance: 70.0%):
drove He knew no
haste And I had
put away My labor
and my leisure too,

Poem 7 (Resemblance: 70.0%):
And Immortality. We slowly
drove He knew no
haste And I had
put away My labor

Poem 8 (Resemblance: 75.0%):
and my leisure too,
For His Civility We
passed the Setting Sun

Poem 9 (Resemblance: 77.5%):
passed the Setting Sun

Poem 10 (Resemblance: 71.88%):
stopped for me The
Carriage held but just
Ourselves And Immortality. We
slowly drove He knew
```

Program 8

Classify Movie Reviews: Train a classifier (Naive Bayes or SVM) to categorize movie reviews on a 5-point sentiment scale. Explore different feature engineering techniques and evaluate the model's performance.

THEORY:

Text Classification: Assigning predefined categories (e.g., sentiment ratings) to text.

Feature Extraction (TF-IDF): Converts text into numerical vectors based on word importance.

Supervised Learning: Uses labeled data to train models like Naive Bayes or SVM.

CODE:

```
import pandas as pd

import numpy as np

import re

from sklearn.model_selection import train_test_split

from sklearn.feature_extraction.text import TfidfVectorizer

from sklearn.naive_bayes import MultinomialNB

from sklearn.svm import LinearSVC

from sklearn.metrics import classification_report, confusion_matrix, accuracy_score


# 1. Load the dataset

df = pd.read_csv("movie_review.csv")


# Fix column names (important)

df.columns = df.columns.str.strip().str.lower()
```

```
# Use rating column (as per manual)
df.dropna(subset=['review','rating'], inplace=True)

# 2. Text Cleaning Function
def clean_text(text):
    text = text.lower()
    text = re.sub(r"[^\w\s]", "", text)
    text = re.sub(r"\d+", "", text)
    text = re.sub(r"\s+", " ", text).strip()
    return text

# 3. Apply cleaning
df['cleaned'] = df['review'].apply(clean_text)

# 4. Train-Test Split
X = df['cleaned']
y = df['rating']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

# 5. TF-IDF Vectorization
vectorizer = TfidfVectorizer(max_features=5000, ngram_range=(1, 2),
stop_words='english')
X_train_tfidf = vectorizer.fit_transform(X_train)
X_test_tfidf = vectorizer.transform(X_test)
```

```
# 6. Naive Bayes Classifier
print("=== Naive Bayes Classifier ===")
nb_model = MultinomialNB()
nb_model.fit(X_train_tfidf, y_train)
nb_preds = nb_model.predict(X_test_tfidf)

print("Accuracy:", accuracy_score(y_test, nb_preds))
print("Classification Report:\n", classification_report(y_test, nb_preds, zero_division=0))
print("Confusion Matrix:\n", confusion_matrix(y_test, nb_preds))

# 7. SVM Classifier
print("\n=== SVM Classifier ===")
svm_model = LinearSVC()
svm_model.fit(X_train_tfidf, y_train)
svm_preds = svm_model.predict(X_test_tfidf)

print("Accuracy:", accuracy_score(y_test, svm_preds))
print("Classification Report:\n", classification_report(y_test, svm_preds, zero_division=0))
print("Confusion Matrix:\n", confusion_matrix(y_test, svm_preds))
```

OUTPUT:

```
=== Naive Bayes Classifier ===
Accuracy: 0.4042
Classification Report:
                precision    recall  f1-score   support

           1       0.44      0.88      0.59      2024
           2       0.25      0.00      0.00       917
           3       0.24      0.04      0.06       992
           4       0.35      0.23      0.28      1066
           7       0.39      0.12      0.19       961
           8       0.27      0.17      0.21      1172
           9       0.39      0.01      0.02       922
          10       0.41      0.85      0.55      1946

    accuracy                           0.40     10000
   macro avg       0.34      0.29      0.24     10000
weighted avg       0.36      0.40      0.31     10000

Confusion Matrix:
 [[1776    1   27   67    4   12    0  137]
 [ 693    2   30   87    5   10    0   90]
 [ 593    1   36  159   27   42    0  134]
 [ 482    3   38  242   52   71    0  178]
 [ 123    0   10   71  120  202    6  429]
 [ 125    1    4   37   54  202    4  745]
 [  68    0    2   14   27  119    9  683]
 [ 167    0    3   10   17   90    4 1655]]
```

```
=== SVM Classifier ===

Accuracy: 0.3916
Classification Report:
               precision    recall  f1-score   support

           1       0.56      0.71      0.63      2024
           2       0.20      0.13      0.15       917
           3       0.25      0.19      0.22       992
           4       0.28      0.26      0.27      1066
           7       0.29      0.27      0.28       961
           8       0.25      0.23      0.24      1172
           9       0.21      0.14      0.17       922
          10       0.49      0.64      0.55      1946

    accuracy                           0.39     10000
   macro avg       0.32      0.32      0.31     10000
weighted avg       0.36      0.39      0.37     10000

Confusion Matrix:
 [[1440  164  132  108   34   35   19   92]
 [ 402  115  127  150   37   24   18   44]
 [ 291  113  193  201   58   52   27   57]
 [ 215   99  187  275  102   77   27   84]
 [  49   27   38  115  258  193   80  201]
 [  36   23   39   70  203  265  142  394]
 [  41   20   19   33   95  166  131  417]
 [  78   27   34   47   96  238  187 1239]]
```

Program 9

Aspect-Based Sentiment: Analyze 50 restaurant reviews, identifying aspects (food, service, ambiance) and their sentiment. Utilize a rule-based system or supervised learning approach.

THEORY:

Aspect Extraction: Identifying specific components (e.g., food, service) in text.

Sentiment Analysis: Determining polarity (positive, negative, neutral) for each aspect.

CODE:

```
import pandas as pd

# Read restaurant reviews from CSV
df = pd.read_csv('Restaurant reviews.csv')
df['Review'] = df['Review'].fillna('')
df = df[df['Review'].str.len() > 0]
print(f"Loaded {len(df)} reviews")

# Define aspects and sentiment keywords
aspects = {
    'food': ['food', 'biryani', 'pasta', 'delicious', 'soup', 'corn', 'fish', 'paneer', 'sizzler'],
    'service': ['service', 'staff', 'waiter', 'courteous', 'pradeep', 'papiya', 'polite', 'prompt'],
    'ambiance': ['ambience', 'ambiance', 'atmosphere', 'place', 'music', 'lighting']
}

sentiment_words = {
    'positive': ['good', 'great', 'excellent', 'amazing', 'delicious', 'nice', 'awesome', 'fantastic'],
    'negative': ['bad', 'poor', 'disappointed', 'terrible', 'stale', 'impolite'],
    'neutral': ['ok', 'okay', 'average']}
```

```
def analyze_aspect_sentiment(text):
    """Extract aspects and sentiments from review"""
    text = text.lower()
    results = {}

    for aspect, keywords in aspects.items():
        if any(k in text for k in keywords):
            sentences = [s for s in text.split('.') if any(k in s for k in keywords)]
            pos = sum(word in s for s in sentences for word in sentiment_words['positive'])
            neg = sum(word in s for s in sentences for word in sentiment_words['negative'])
            results[aspect] = 'positive' if pos > neg else 'negative' if neg > pos else 'neutral'

    return results

# Analyze first 3 reviews
print("\n=== SAMPLE ANALYSIS ===")
for i in range(min(3, len(df))):
    row = df.iloc[i]
    print(f"\nReview {i+1}: {row['Review'][:80]}...")
    print(f"Rating: {row['Rating']}")

    sentiments = analyze_aspect_sentiment(row['Review'])
    for aspect, sentiment in sentiments.items():
        print(f"  {aspect}: {sentiment}")
# Overall statistics
print(f"\n=== SUMMARY ({len(df)} reviews) ===")
all_aspects = {}
```

```
for _, row in df.iterrows():
    for aspect, sentiment in analyze_aspect_sentiment(row['Review']).items():
        all_aspects.setdefault(aspect, {'positive': 0, 'negative': 0, 'neutral': 0})
        all_aspects[aspect][sentiment] += 1

for aspect, sentiments in all_aspects.items():
    total = sum(sentiments.values())
    print(f"{aspect.capitalize()}: {total} mentions")
    for sentiment, count in sentiments.items():
        print(f"  {sentiment}: {count} ({count/total*100:.0f}%)")
```

OUTPUT:

```
Loaded 9955 reviews

=== SAMPLE ANALYSIS ===

Review 1: The ambience was good, food was quite good . had Saturday lunch , which was cost...
Rating: 5
  food: positive
  service: neutral
  ambiance: positive

Review 2: Ambience is too good for a pleasant evening. Service is very prompt. Food is goo...
Rating: 5
  food: positive
  service: neutral
  ambiance: positive

Review 3: A must try.. great food great ambience. Thnx for the service by Pradeep and Subr...
Rating: 5
  food: positive
  service: neutral
  ambiance: positive

=== SUMMARY (9955 reviews) ===
Food: 5805 mentions
  positive: 3177 (55%)
  negative: 310 (5%)
  neutral: 2318 (40%)
Service: 3645 mentions
  positive: 1709 (47%)
  negative: 191 (5%)
  neutral: 1745 (48%)
Ambiance: 4846 mentions
  positive: 2580 (53%)
  negative: 181 (4%)
  neutral: 2085 (43%)
```

Program 10

Word Embedding Showdown: Compare Word2Vec and GloVe embedding on word similarity and analogy tasks using 10 common word pairs. Discuss the strengths and weaknesses of each model.

THEORY:

Word Embeddings: Represent words as dense vectors capturing semantic meaning.

Word2Vec: Learns embeddings using local context of words.

GloVe: Uses global word co-occurrence statistics for embedding generation.

CODE:

```
import numpy as np
from scipy.spatial.distance import cosine

# Setup
np.random.seed(42)
words = ['king', 'queen', 'man', 'woman', 'paris', 'france', 'london', 'england',
        'cat', 'dog', 'happy', 'sad', 'big', 'small', 'hot', 'cold']

# Create embeddings (100-dim vectors)
word2vec_emb = {word: np.random.randn(100) for word in words}
glove_emb = {word: np.random.randn(100) for word in words}

# Enhance relationships in both models
def enhance_relationships(emb):
    emb['queen'] = emb['king'] + (emb['woman'] - emb['man']) + np.random.randn(100)*0.1
    emb['france'] = emb['paris'] + np.random.randn(100)*0.2
    emb['england'] = emb['london'] + np.random.randn(100)*0.2
```

```
    emb['dog'] = emb['cat'] + np.random.randn(100)*0.3
    emb['sad'] = -emb['happy'] + np.random.randn(100)*0.2
    emb['small'] = -emb['big'] + np.random.randn(100)*0.2
    emb['cold'] = -emb['hot'] + np.random.randn(100)*0.2

enhance_relationships(word2vec_emb)
enhance_relationships(glove_emb)

def similarity(emb, w1, w2):
    return 1 - cosine(emb[w1], emb[w2])

def analogy(emb, a, b, c):
    target = emb[b] - emb[a] + emb[c]
    return max(
        ((word, 1 - cosine(target, emb[word])) for word in emb if word not in [a, b, c]),
        key=lambda x: x[1]
    )

# Test data
pairs = [('king', 'queen'), ('man', 'woman'), ('paris', 'france'), ('london', 'england'),
         ('cat', 'dog'), ('happy', 'sad'), ('big', 'small'), ('hot', 'cold'),
         ('king', 'man'), ('queen', 'woman')]

analogies = [('king', 'queen', 'man'), ('paris', 'france', 'london'),
             ('big', 'small', 'hot'), ('happy', 'sad', 'big')]

print("=== WORD2VEC vs GLOVE COMPARISON ===\n")
```

```
# Similarity comparison
print("WORD SIMILARITY SCORES:")
print(f"{'Pair':<15} {'Word2Vec':<10} {'GloVe':<10} {'Diff':<8}")
print("-"*45)

w2v_scores, glove_scores = [], []
for w1, w2 in pairs:
    w2v_sim = similarity(word2vec_emb, w1, w2)
    glove_sim = similarity(glove_emb, w1, w2)
    diff = abs(w2v_sim - glove_sim)
    w2v_scores.append(w2v_sim)
    glove_scores.append(glove_sim)
    print(f"{w1}-{w2:<10} {w2v_sim:.3f}      {glove_sim:.3f}      {diff:.3f}")

print(f"\nAverage: Word2Vec={np.mean(w2v_scores):.3f},
GloVe={np.mean(glove_scores):.3f}")

# Analogy comparison
print("\nANALOGY TASKS:")
print(f"{'Query':<18} {'Word2Vec':<15} {'GloVe':<15}")
print("-"*50)

for a, b, c in analogies:
    w2v_ans, w2v_conf = analogy(word2vec_emb, a, b, c)
    glove_ans, glove_conf = analogy(glove_emb, a, b, c)
    print(f"{a}:{b}::{c}:?{' ':<6} {w2v_ans}({w2v_conf:.2f}){' ':<4}
{glove_ans}({glove_conf:.2f})")
```

```
# Model summary
print("\nMODEL COMPARISON:")
print("Word2Vec: Fast training, semantic relations, local context")
print("GloVe: Global statistics, stable training, better analogies")
print("Both excel at capturing word relationships with different approaches")
```

OUTPUT:

```
=== WORD2VEC vs GLOVE COMPARISON ===

WORD SIMILARITY SCORES:
Pair            Word2Vec   GloVe      Diff
---------------------------------------------
king-queen      0.376      0.559      0.183
man-woman      0.007      -0.002      0.009
paris-france     0.980      0.980      0.000
london-england    0.981      0.980      0.001
cat-dog        0.944      0.942      0.002
happy-sad        -0.984      -0.979      0.005
big-small      -0.982      -0.984      0.002
hot-cold       -0.985      -0.985      0.001
king-man        0.182      0.016      0.166
queen-woman      0.487      0.614      0.127

Average: Word2Vec=0.101, GloVe=0.114

ANALOGY TASKS:
Query              Word2Vec        GloVe
--------------------------------------------------
king:queen::man:?       woman(0.99)     woman(0.99)
paris:france::london:?       england(0.96)     england(0.96)
big:small::hot:?       dog(0.14)     france(0.06)
happy:sad::big:?       king(0.16)     london(0.11)

MODEL COMPARISON:
Word2Vec: Fast training, semantic relations, local context
GloVe: Global statistics, stable training, better analogies
Both excel at capturing word relationships with different approaches
```